

☰ Hide menu

Week 5: Image Processing I

✔ **Reading:** Week 5 Lecture Handout
30 min

✔ **Ungraded Plugin:** 5.1 Overview of Image Processing I
4 min

✔ **Ungraded Plugin:** 5.2 Pixel Processing
3 min

✔ **Practice Quiz:** 5.2 Pixel Processing Self-check Quiz
Started

✔ **Reading:** 5.3 Video Correction
10 min

✔ **Ungraded Plugin:** 5.3 LSIS and Convolution
22 min

✔ **Practice Quiz:** 5.3 LSIS and Convolution Self-check Quiz
Started

✔ **Reading:** 5.4 Video Correction
10 min

✔ **Ungraded Plugin:** 5.4 Linear Image Filters
16 min

✔ **Practice Quiz:** 5.4 Linear Image Filters Self-check Quiz
Started

✔ **Ungraded Plugin:** 5.5 Non-Linear Image Filters
16 min

✔ **Practice Quiz:** 5.5 Non-Linear Image Filters Self-check Quiz
Started

✔ **Reading:** 5.6 Video Correction
10 min

🔗 **Ungraded Plugin:** 5.6 Template Matching by Correlation
7 min

📋 **Practice Quiz:** 5.6 Template Matching by Correlation Self-check Quiz
10 min

💬 **Discussion Prompt:** Week 5 Analyzing Convolution
10 min

💬 **Discussion Prompt:** Week 5 Questions and Feedback

Week 5 Quiz

5.6 Video Correction

In [Video 5.6 Template Matching by Correlation](#) ↗, from 2:18 to 3:10, adding "Convolution = Correlation when Template is Symmetric" (see below).

*The revisions are shown in **red** and correspond to either corrections to errors or additions made for clarity.

< 1 / 1 >

🔍 100% 🔍

⬇️ ⚙️

Convolution vs. Correlation

Convolution:

$$g[i,j] = \sum_m \sum_n f[m,n] \underline{t[i-m,j-n]} = t * f$$

Correlation:

$$R_{tf}[i,j] = \sum_m \sum_n f[m,n] \underline{t[m-i,n-j]} = t \otimes f$$

No Flipping in Correlation

Convolution = Correlation when Template is Symmetric

Go to next item ✓ Completed

👍 Like 🗨 Dislike 🚩 Report an issue